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Aim :Implementing Web Scraping in Python with BeautifulSoup.

1. What is web scraping? Explain the role of the Requests library in web scraping.
Web scraping is the process of automatically extracting data from websites using programming. Instead of manually copying data, a script collects the required information quickly and efficiently. The Requests library plays an important role in web scraping as it is used to send HTTP requests to a website and retrieve its HTML content. It acts like a web browser and helps in downloading the webpage data, which is then passed to other tools like BeautifulSoup for further processing.
2. What is BeautifulSoup? How does it help in parsing HTML? BeautifulSoup is a Python library used for parsing HTML and XML documents. It helps convert raw HTML code into a structured tree format, making it easier to navigate and extract useful data. With BeautifulSoup, users can search for specific tags, attributes, and text within a webpage. It simplifies the process of handling messy or complex HTML and allows easy extraction of information such as headings, links, and paragraphs.
3. Differentiate between web scraping and web crawling. Web scraping and web crawling are related but different concepts. Web scraping focuses on extracting specific data from a webpage, such as quotes, prices, or text. On the other hand, web crawling is the process of browsing and indexing multiple web pages automatically by following links. Crawling is used by search engines to discover content on the internet, while scraping is mainly used to collect particular information from selected pages.
4. What are the commonly used methods in BeautifulSoup? BeautifulSoup provides several useful methods to extract and manipulate data from HTML. The `find()` method is used to locate the first matching tag, while `find_all()` retrieves all matching elements. The `get_text()` method extracts only the text content from HTML tags, and the `get()` method is used to fetch attribute values like links. Additionally, the `select()` method allows searching using CSS selectors. These methods make it easy to navigate and extract specific data from web pages.
5. What are the ethical and legal considerations in web scraping? Web scraping must be done carefully by following ethical and legal guidelines. Users should

always check the website's robots.txt file to see if scraping is allowed. It is important not to overload the server by sending too many requests in a short time. Scraping private or sensitive data should be avoided, and the website's terms and conditions must be respected. Giving proper credit and using data responsibly ensures that scraping is done in a fair and legal manner.

```
# Step 0: Install required libraries
!pip install requests beautifulsoup4 pandas
```

```
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages
```

```
# Step 1: Import libraries
import requests
from bs4 import BeautifulSoup
import pandas as pd
```

```
# Step 2: Send HTTP request
url = "http://books.toscrape.com/"
response = requests.get(url)
```

```
# Step 3: Parse HTML content
soup = BeautifulSoup(response.text, "html.parser")
```

```
# Step 4: Extract book data
books_data = []

books = soup.find_all("article", class_="product_pod")

for b in books:
    title = b.h3.a["title"]
    price = b.find("p", class_="price_color").get_text()
    availability = b.find("p", class_="instock availability").get_text().strip()

    books_data.append({
        "Title": title,
```

```
"Price": price,  
"Availability": availability  
})
```

```
# Step 5: Convert to DataFrame  
df = pd.DataFrame(books_data)
```

```
# Step 6: Display data  
print(df.head())
```

	Title	Price	Availability
0	A Light in the Attic	Â£51.77	In stock
1	Tipping the Velvet	Â£53.74	In stock
2	Soumission	Â£50.10	In stock
3	Sharp Objects	Â£47.82	In stock
4	Sapiens: A Brief History of Humankind	Â£54.23	In stock

```
# Step 7: Save to CSV  
df.to_csv("books_scraped.csv", index=False)
```